

MALAMA MASHEKE

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EDUCATION

Livingstone College

B.S. Computer Information Systems

Expected Graduation: December 2029

GPA: 4.0/4.0

Relevant Coursework: Computer Systems, Intro to Programming, Calculus, Intro to Computer Science (Harvard CS50), Python: Machine Learning and Deep Learning (Udemy)

TECHNICAL PROJECTS

Demos | Voting Platform with Data-Layer Authorization

July 2026

demos-rust-eight.vercel.app

- A live web platform that lets an organization run group votes, from a simple poll to a ranked-choice election, and get a result everyone can trust.
- Defined the requirements and data model with one collaborator, then built the entire backend while he built the interface, covering database design, user accounts and permissions, and the vote-counting logic including ranked-choice tallying that eliminates the lowest option round by round.
- Structured the data so single-choice, multiple-choice, and future weighted voting all run on one design, and made duplicate or contradictory ballots impossible to record rather than filtering them out afterward.
- Audited my own design, found that two parts of the database could be cross-referenced to work out how an individual had voted, and closed it without breaking the ranked-choice count that relied on the link I removed.
- Deployed and in use. Access rules live in the database rather than the application, so no future feature can quietly bypass them, and results stay accurate because every tally is recalculated from the source data rather than stored and trusted.
- Recommended turning off live results for ranked-choice votes after finding standings can reverse between rounds, choosing a less impressive display over one that would mislead people mid-vote.
- **Technical Infrastructure:** PostgreSQL, Data Modeling, Access Control and Authorization Design, API Design, TypeScript, Next.js

Tri-Ask-Me | AI Support Triage Agent

May 2026

- An AI assistant that reads incoming customer support tickets and decides what each one is about, how urgent it is, and who should handle it.
- Built in Python over 24 hours and tested against real tickets from three different products. It looks up the relevant help documentation before replying, so answers come from real sources, and it declines to answer at all when it finds nothing relevant.
- Designed it as four separate stages, each handling one job, and made every decision carry a category, urgency level, destination, and written reason, so results are reviewable by someone who has never seen the code.
- Cuts the manual effort of sorting a support queue while routing anything sensitive, such as billing, legal, security, or account access, to a person rather than guessing. Placed in the top 10% globally out of over 15,000 participants.
- **Technical Infrastructure:** Python, Amazon Bedrock, Retrieval-Augmented Generation, TF-IDF, Prompt Engineering

Spectrum Sizzle Workshop | Baylor University, SMART Hub

June 2026

- One of 40 undergraduates selected nationwide for an NSF-funded research workshop on wireless communication and how the airwaves are shared and regulated, led by Google and Keysight engineers.
- Analyzed measurements captured on professional test equipment using MATLAB, examining how signals are encoded and how limited frequency space gets allocated between competing users.
- Covered spectrum policy, wireless circuit design, and radar systems, linking engineering constraints to the regulations that decide what can be built.
- Turned raw instrument readings into conclusions I could defend against theory, presenting and comparing findings with students from institutions across the country.
- **Technical Infrastructure:** MATLAB, Data Analysis and Interpretation, Signal Processing, Frequency Allocation Analysis, Communication Systems and Modulation, Radar Systems

COMMUNITIES & LEADERSHIP

- **Adobe Student Ambassador:** Selected to represent Adobe on campus. Run peer education sessions and live product demonstrations, translating complex creative tools into plain guidance for students with no technical background. Track where users get stuck, rework how those steps are taught, and gather feedback that has driven adoption across campus.

ADDITIONAL SKILLS

- **Languages:** Python, C, JavaScript/TypeScript, SQL, MATLAB
- **Tools:** Codex, Claude Code, Git, GitHub, Bash, VS Code, Jupyter Notebook, Google Colab
- **Foundations:** API Design, Client-Server Architecture, Query Design, Machine Learning and Deep Learning

AWARDS & RECOGNITION

- **HackerRank Worldwide Hackathon:** Top 10% Globally out of over 15,000 participants
- **Odessa J. Robinson Scholarship:** Promise City Church
- **Lwengu School, Monze, Zambia (Jan 2019 - Nov 2021):** Class Salutatorian | Overall Best in Mathematics | Leadership Award